

Name: Octavio MoralesDate: November 3, 2022Pledge: I pledge my honor that I have abided by the Stevens Honor System.

1) Consider an array containing the following 40 integers:

5 2 4 4 0 1 6 7 3 1 1 0 5 1 5 4 4 5 7 0 6 1 0 7 5 2 7 6 5 3 7 0 5 5 7 1 1 2 6 5

How many counters does CountingSort need to sort this array: (In image below)

Give the value of each counter after the array of counters has been fully initialized:

(In image below)

Octavio Morales
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1. 5 2 4 4 0 1 6 7 3 1 1 0 5 1 5 4 4 5 7 0 6 1 0 7 5 2 7 6 5 3 7 0 5 5 7 1 1 2 6 5

① Counter

0	1	2	3	4	5	6	7
5	7	3	2	4	9	4	6

② Cumulation

0	1	2	3	4	5	6	7
5	12	15	17	21	30	34	40

③ Shift

0	1	2	3	4	5	6	7
0	5	12	15	17	21	30	34

Counters: 8

Sorted Array

0 0 0 0 0 1 1 1 1 1 1 2 2 2 3 3 4 4 4 4 5 5
5 5 5 5 5 5 5 6 6 6 6 7 7 7 7 7

- 2) Consider an array containing the following 32 bit integers (written as hexadecimal values to save space):

4EC1EEA9
 520B6E78
 1E90D74E
 52DB6E42
 5F05EF13
 74284442
 794E8117
 55526E42

Imagine you are using a version of RadixSort that sorts on one byte at a time (so two hexadecimal digits) using a stable version of CountingSort. Write the content of the array after each of the four runs of CountingSort:

Base	Run 1	Run 2	Run 3
2. 4EC1EEA9	5F05EF13	74284442	5F05EF13
520B6E78	794E8117	52DB6E42	520B6E78
1E90D74E	52DB6E42	55526E42	74284442
52DB6E42	74284442	520B6E78	794E8117
5F05EF13	55526E42	794E8117	55526E42
74284442	1E90D74E	1E90D74E	1E90D74E
794E8117	520B6E78	4EC1EEA9	4EC1EEA9
55526E42	4EC1EEA9	5F05EF13	52DB6E42
Run 4			
1E90D74E			
4EC1EEA9			
520B6E78			
52DB6E42			
55526E42			
5F05EF13			
74284442			
794E8117			